Intervention Analysis on the Effectiveness of Plastic Waste Reduction Policies in Palu City

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Abstract*—This study aims to quantitatively evaluate the effectiveness of the plastic waste reduction policy, specifically Palu Mayor Regulation No. 40 of 2021. The methodology employed is Intervention Analysis on monthly waste volume data from January 2015 to December 2025 to measure the significant impact of the policy after controlling for existing trend and seasonal patterns. The data were analyzed using a seasonal ARIMA model with a step function intervention variable. Results indicate that the ARIMA(0,1,1)(1,1,1)$_{12}$ model with intervention component is the most optimal specification with the lowest AIC value (1,321.99) and MAPE of 1.13%. The intervention analysis identified a statistically significant structural change following policy implementation in March 2022, with an intervention effect of +170.40 tons per month (p = 0.0253). This counterintuitive finding indicates a 2.91% increase in recorded waste volume compared to the pre-intervention baseline, which may be attributed to improved waste collection efficiency or better reporting coverage. Residual diagnostics confirm model adequacy with white noise residuals and approximate normality. The implications of these findings can serve as a basis for strengthening environmental policy in Palu City and as a methodological reference for other cities in evaluating the effectiveness of waste management programs.

***Keywords*—Environmental Policy, Intervention Analysis, Plastic Waste Management, Seasonal ARIMA, Time Series**

I. INTRODUCTION

Plastic pollution has emerged as a critical environmental issue worldwide, particularly in urban centers where consumption of single-use plastics continues to grow rapidly [1]. The durability and low degradability of plastics contribute to their persistence in the environment, leading to soil and marine contamination, biodiversity loss, and long-term ecological harm [2]. According to the World Bank, plastic waste constitutes nearly 12% of total municipal solid waste in developing countries, with a significant portion generated from packaging and disposable consumer goods [3]. Indonesia ranks among the largest contributors to global marine plastic pollution, generating approximately 3.2 million tons of plastic waste annually, of which 1.3 million tons are mismanaged [4]. Rapid urbanization and weak waste management systems have exacerbated the issue, creating challenges for local governments in maintaining sustainable urban environments [5]. In response, the Indonesian government has introduced several national strategies, including the Presidential Regulation No. 97 of 2017 on waste management and the National Action Plan for Marine Debris Reduction (2018-2025), aiming to reduce plastic waste leakage by 70% by 2025 [6].

At the municipal level, Palu City has also taken regulatory action to address the growing problem of single-use plastic waste through the issuance of Mayoral Regulation No. 40 of 2021. This regulation mandates the reduction of plastic bags, straws, and other disposable plastics in retail outlets, restaurants, and public facilities [7]. The policy represents a local implementation of the national commitment to achieve sustainable waste reduction and promote a circular economy. However, as with many local initiatives, the effectiveness of such a regulation depends on its practical enforcement and behavioral compliance within the community [8]. Despite the regulation's enactment, the total volume of plastic waste in Palu City remains relatively high, indicating potential challenges in behavioral adaptation and enforcement consistency. While environmental campaigns have increased public awareness, quantitative evidence assessing the real effect of the policy remains limited. Understanding whether the Mayoral Regulation No. 40 of 2021 has led to a measurable decline in plastic waste generation is therefore essential for evaluating its effectiveness and guiding future policy adjustments [3].

Intervention analysis provides a robust quantitative framework to measure the impact of a policy or event on time-dependent data [9]. By incorporating a step-type intervention variable, the method allows researchers to assess whether a structural change has occurred following the implementation of a policy. This analytical framework has been widely applied in various fields such as economics, epidemiology, and environmental studies to evaluate the effectiveness of interventions and policy shocks [10], [11]. Therefore, this research applies an intervention analysis model to evaluate the effectiveness of Mayoral Regulation No. 40 of 2021 in reducing plastic waste generation in Palu City. Using monthly plastic waste data from January 2015 to December 2025, the study aims to: (1) develop and identify the optimal intervention model for the waste generation data of Palu City, accounting for the impact of the regulation; and (2) analyze and quantify the significance and magnitude of the impact of implementing the regulation as a policy intervention on the waste volume trend in Palu City.

II. LITERATURE REVIEW

A. Time Series Analysis

A time series $\{Y_t\}$ is a sequence of random variables indexed by time t, recorded at equally spaced intervals [10]. The observations may display temporal dependence, meaning that past values influence current or future observations. A time series can be decomposed into its components as shown in Equation (1):

$$Y_t = T_t + S_t + C_t + \varepsilon_t \tag{1}$$

where Y_t is the observed value at time t, T_t is the trend component, S_t is the seasonal component, C_t is the cyclical component, and ε_t is the random error term. Stationarity is a fundamental assumption in time series analysis, where a stationary series has constant mean, variance, and autocovariance over time [10]. The Augmented Dickey-Fuller (ADF) test is commonly used to examine stationarity, expressed as:

$$\Delta Y_t = \alpha + \beta t + \gamma Y_{t-1} + \sum_{i=1}^{p} \delta_i \Delta Y_{t-i} + \varepsilon_t \tag{2}$$

where ΔY_t is the first-differenced series, γ is the unit root coefficient, α is the intercept, and βt is the deterministic time trend [12].

B. ARIMA Model

The Autoregressive Integrated Moving Average (ARIMA) model, developed by Box and Jenkins [13], integrates three major components: the Autoregressive (AR) part that captures dependence on past values, the Integrated (I) part that accounts for differencing to achieve stationarity, and the Moving Average (MA) part that models serial correlation in the residuals. The general expression for an ARIMA(p,d,q) model is given by:

$$\Phi(B)(1-B)^d Y_t = \Theta(B)\varepsilon_t \tag{3}$$

where Y_t is the observed value at time t, B is the backshift operator where $BY_t = Y_{t-1}$, d is the order of non-seasonal differencing, $\Phi(B)$ is the autoregressive polynomial of order p, $\Theta(B)$ is the moving average polynomial of order q, and ε_t is the white noise error term. The AR(p) process is defined as:

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + \varepsilon_t \tag{4}$$

The MA(q) process is expressed as:

$$Y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \cdots + \theta_q \varepsilon_{t-q} \tag{5}$$

When the data exhibit periodic or seasonal fluctuations, the model can be extended into a Seasonal ARIMA (SARIMA) form, denoted as ARIMA(p,d,q)$(P,D,Q)_s$, where s indicates the seasonal period [10].

C. Intervention Analysis (Box-Tiao Model)

Intervention analysis, introduced by Box and Tiao [9], extends the ARIMA framework by incorporating external shocks or interventions as explanatory variables. The general intervention model can be expressed as:

$$Y_t = \frac{\omega(B)}{\delta(B)} X_t + N_t, \quad N_t = \frac{\theta(B)}{\phi(B)} \varepsilon_t \tag{6}$$

where Y_t is the observed time series, X_t is the intervention variable at time t, $\omega(B)$ is the numerator polynomial representing intervention impact magnitude, $\delta(B)$ is the denominator polynomial representing dynamic decay, B is the backshift operator, and N_t is the noise component modeled as an ARIMA process. A step function models a permanent change that begins at the time of intervention and persists indefinitely [10]:

$$S_t = 0, if\, t < T; 1, if\, t \geq T \tag{7}$$

A pulse function models a temporary or short-lived change that occurs only in a single period (or a few consecutive periods) [10]:

$$P_t = 0, if\, t = T; 1, if\, t \neq T \tag{8}$$

D. Model Diagnostic and Accuracy

Residual independence can be tested using the Ljung-Box Q-statistic, defined as [14]:

$$Q = n(n+2) \sum_{k=1}^{h} \frac{r_k^2}{n-k} \tag{9}$$

where Q is the Ljung-Box test statistic, n is sample size, r_k is the autocorrelation at lag k, and h is the number of lags tested. Model selection is performed using the Akaike Information Criterion (AIC) and Schwarz Bayesian Criterion (SBC) [15]:

$$AIC = -2\, ln(L) + 2k \tag{10}$$

$$SBC = -2\, ln(L) + k\, ln(n) \tag{11}$$

Model accuracy is evaluated using Mean Absolute Percentage Error (MAPE), Root Mean Square Error (RMSE), and Mean Percentage Error (MPE):

$$MAPE = \frac{100}{n} \sum_{t=1}^{n} \left| \frac{Y_t - \hat{Y}_t}{Y_t} \right| \tag{12}$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^{n} \left(Y_t - \hat{Y}_t\right)^2} \tag{13}$$

$$MPE = (1/n) \sum \left(\left(Y_t - \hat{Y}_t\right) / Y_t \right) \times 100\% \tag{14}$$

III. METHODOLOGY

This study utilizes secondary data obtained from the Environmental Agency (DLH) of Palu City. The dataset consists of 128 monthly observations of plastic waste volume (in tons) from January 2015 to December 2025. The dependent variable is the monthly plastic waste volume, while the independent intervention variable is specified as a binary step function where $X_t = 0$ for the pre-intervention period (January 2015 – February 2022) and $X_t = 1$ for the post-intervention period (March 2022 – December 2025).

The analytical procedure follows the Box-Jenkins methodology: (1) exploratory analysis and stationarity assessment using ADF test; (2) model identification through ACF and PACF analysis; (3) parameter estimation using Maximum Likelihood Estimation (MLE); (4) diagnostic checking using Ljung-Box Q-test and normality tests; (5) intervention model specification and estimation; and (6) model validation and forecasting. All analyses were performed using SAS software.

IV. RESULTS AND DISCUSSION

A. Data Characteristics and Time Series Visualization

Table 1 presents the descriptive statistics of plastic waste volume for the entire observation period. The overall mean plastic waste volume is 5,855.64 tons per month with a standard deviation of 230.10 tons. A notable difference exists between the pre-intervention period (mean = 5,775tons) and post-intervention period (mean = 6,019 tons), suggesting a potential increase in waste volume following policy implementation.

Table 1.
Descriptive Statistics of Plastic Waste Volume (tons)

Period	N	Mean	Std Dev	Min	Max
Overall	128	5,856	230.10	5,195.68	6,291.14
Pre-intervention	86	5,776	202.09	5,195.68	6,291.14
Post-intervention	42	6,019	196.50	5,425.00	6,291.14

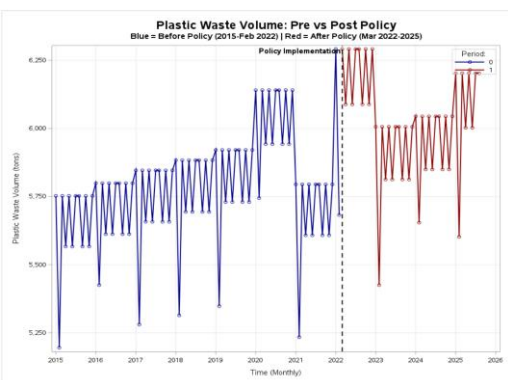

Figure 1. Time Series of Plastic Waste Volume in Palu City (January 2015 - December 2025)

Figure 1 displays the time series plot of plastic waste volume from January 2015 to December 2025. The vertical black dashed line indicates the intervention point (March 2022) when the plastic waste management policy was implemented. Visual inspection reveals a clear seasonal pattern with a period of 12 months, indicating annual cyclical behavior in plastic waste generation. There appears to be a level shift following the intervention point, with generally higher values observed in the post-intervention period.

B. Box-Plot Analysis

The comparative box-plot analysis in Figure 2 directly contrasts the distribution of plastic waste volume between the pre-policy period (n = 86 observations) and the post-policy period (n = 42 observations). The post-policy box-plot demonstrates an elevated position relative to the pre-policy period, with the entire interquartile range positioned above the median of the pre-policy distribution.

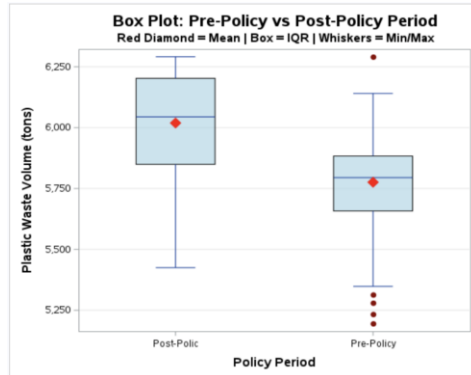

Figure 2. Box Plot Comparison: Pre-Policy (2015-Feb 2022) vs Post-Policy (Mar 2022-Dec 2025)

This represents an increase of approximately 243.24 tons in the average monthly plastic waste volume following policy implementation. The independent samples t-test confirms that this mean difference is statistically significant ($t = -6.45$, $df = 126$, $p < 0.0001$), as shown in Table 2.

Table 2.
Independent Samples T-Test Results

Method	Variance	df	t-Value	p-value	95% CI
Pooled	Equal	126	-6.45	<0.0001	[-317.8, -168.6]
Satterthwaite	Unequal	83	-6.51	<0.0001 ***	[-317.5, -169.0]

C. Stationarity Assesment

The ADF test was applied to examine stationarity under the hypotheses: H_0: The time series contains a unit root (non-stationary); H_1: The time series does not contain a unit root (stationary). Results in Table 3 show that the ADF test strongly rejects the null hypothesis at the 1% significance level after first-order differencing, confirming that the first differenced series is stationary.

Table 3.
ADF Test Results on First Differenced Data

Type	Lag	Rho	Tau	Pr < Tau	Decision
Zero Mean	1	-464.17	-15.22	<0.0001*	Stationary
Single Mean	1	-466.86	-15.20	<0.0001*	Stationary
Trend	1	-466.89	-15.14	<0.0001*	Stationary

D. Model Iddentification and Selection

Based on ACF and PACF analysis of the differenced series shown in Figure 3, four candidate models were proposed. The ACF plot showed a significant spike at lag 1 and a strong

seasonal spike at lag 12, while the PACF also indicated notable correlations at these lags.

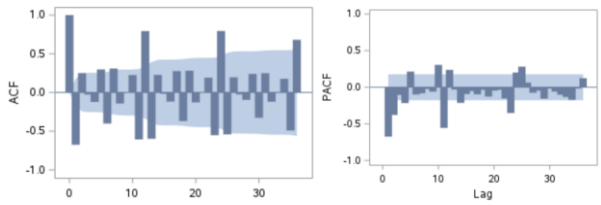

Figure 3. ACF (left) and PACF (right) of First Differenced Plastic Waste Volume

Table 4 presents the comparison of candidate models based on AIC and SBC values. Model 3: ARIMA(0,1,1)(1,1,1)$_{12}$ exhibits the lowest AIC (1,321.99) and SBC (1,330.22) values, providing the best balance between goodness-of-fit and model parsimony.

Table 4.
Comparison of Candidate ARIMA Models

Model	AIC	SBC	ΔAIC	Evidence
ARIMA(0,1,1)(1,1,1)$_{12}$	1,321.9	1,330.2	0.00	Substantial
ARIMA(0,1,1)(2,0,0)$_{12}$	1,528.2	1,536.8	206.2	No support
ARIMA(0,1,[1,6])(2,0,0)$_{12}$	1,561.9	1,573.3	239.9	No support
ARIMA(0,1,1)(1,0,0)$_{12}$	1,623.2	1,628.9	301.2	No support

E. Parameter Estimation

The parameters of the selected ARIMA(0,1,1)(1,1,1)$_{12}$ model were estimated using Maximum Likelihood Estimation. Table 5 presents the parameter estimates, where all parameters are statistically significant at the 5% level, indicating that the model adequately captures the temporal structure of the data.

Table 5. Parameter Estimates for ARIMA(0,1,1)(1,1,1)$_{12}$ Model

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Parameter Estimates for ARIMA(0,1,1)(1,1,1)$_{12}$ Model

Parameter	Estimate	Std Error	t-value	p-value	Lag
$MA_{1,1}$	0.2505	0.1246	2.01	0.0444*	1
$MA_{1,2}$ (Seasonal)	0.7495	0.1573	4.77	<0.0001***	12
$AR_{1,1}$ (Seasonal)	-0.8756	0.0517	-16.94	<0.0001***	12

F. Intervention Analysis

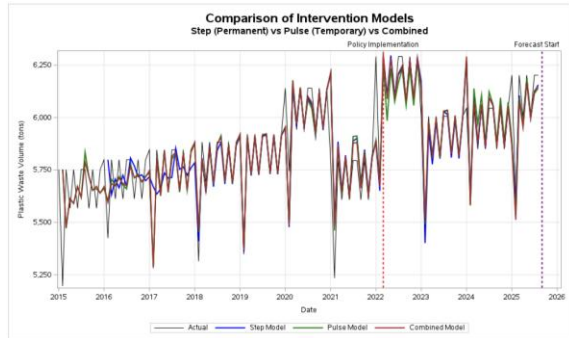

Figure 4. Visual Comparison of Step, Pulse, and Combined Intervention Models

The step intervention model achieved the lowest AIC (1,421.66) and SBC (1,440.87) values, substantially outperforming all alternative specifications. The step intervention parameter (ω_0 = 170.40, SE = 75.21, t = 2.27, p = 0.0255) is statistically significant at the α = 0.05 level, indicating a permanent increase of approximately 170.40 tons in the monthly plastic waste volume following policy implementation. Table 6 presents the intervention model comparison and Table 7 summarizes the intervention effect findings.

Table 6.
Intervention Model Comparison

Model	AIC	SBC	ΔAIC	Evidence
Step (Permanent)	1,421.66	1,440.87	0.00	Substantial
Combined (Step+Pulse)	1,560.65	1,577.72	138.99	No support
Pulse (Temporary)	1,562.03	1,576.25	140.37	No support

Table 7.
Summary of Intervention Effect Findings

Metric	Value	Interpretation
Intervention Effect (ω_0)	+170.40 tons	Significant increase
95% Confidence Interval	[21.49, 319.31]	Excludes zero
Percentage Change	+2.91%	Relative to pre-policy mean
Effect Type	Permanent (Step)	Sustained level shift
Statistical Significance	p = 0.0253	Significant at α = 0.05

G. Diagnostic Checking

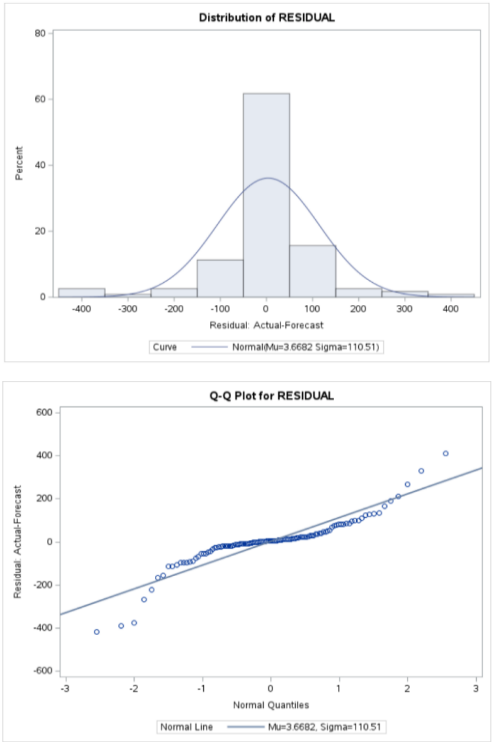

Figure 5. Residual Distribution Histogram (top) and Q-Q Plot (bottom)

The residual histogram and Q–Q plot indicate that the residuals are approximately normally distributed, with most observations closely following the theoretical normal line and only minor deviations at the extreme tails. This graphical evidence is consistent with the Ljung–Box Q test results in Table 8, which show no significant residual autocorrelation at lag 12 and above. Together, these diagnostics confirm that the residuals behave as independent and approximately normal white noise, satisfying the key assumptions required for valid statistical inference.

Table 8.
Ljung-Box Q Test for Residual Autocorrelation

To Lag	Chi-Square	DF	Pr > ChiSq	Decision
6	2.80	6	0.8341	White noise ✓
12	21.34	12	0.0457	Marginal
18	27.33	18	0.0730	White noise ✓
24	28.91	24	0.2237	White noise ✓

Table 8 reports the results of the Ljung–Box Q test applied to the residuals of the fitted Seasonal ARIMA intervention model across multiple lag lengths. Overall, the test results indicate that the residuals behave as *white noise*, particularly for lag orders 12 and above, which are critical in the context of monthly time series data.

At lag 6, the Ljung–Box test yields a p-value of 0.8341**,** providing strong evidence to fail to reject the null hypothesis of no residual autocorrelation. At lag 12, the p-value of 0.0457 lies close to the conventional significance threshold, suggesting a marginal indication of residual dependence. However, this result should be interpreted with caution, as isolated marginal significance at a single lag is commonly observed in finite samples and does not necessarily imply model inadequacy.

Importantly, for higher lag orders (18 and 24), the p-values increase to 0.0730 and 0.2237, respectively, both exceeding the 5% significance level. These results indicate that no systematic autocorrelation persists in the residuals at longer lags. The consistency of non-significant Ljung–Box statistics at lag 12 and beyond supports the conclusion that the residuals approximate *white noise*, implying that the model has adequately captured the underlying temporal structure of the data.

Taken together, the Ljung–Box Q test results confirm that the fitted model satisfies the residual independence assumption required for valid statistical inference and reliable policy impact evaluation.

H. Model Performance and Actual vs Fitted Values

Figure 6 illustrates the comparison between actual plastic waste volume and fitted values obtained from the final ARIMA model with intervention. The model demonstrates excellent fit with MAPE of 1.13% as shown in Table 9.

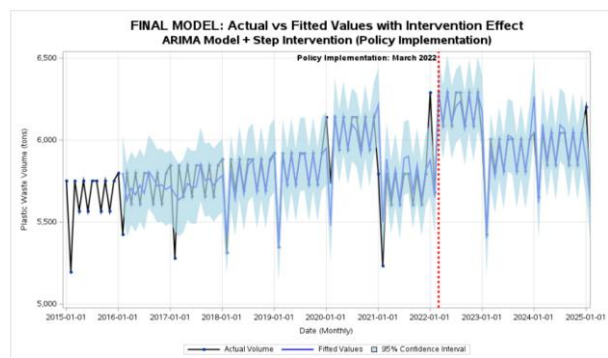

Figure 6. Actual vs Fitted Values with Intervention Effect

Table 9.
Model Accuracy Metrics

MAE (tons)	MAPE (%)	MPE (%)	RMSE (tons)	Rating
65.41	1.13	0.03	110.09	Excellent

Table 9 presents the accuracy metrics of the proposed Seasonal ARIMA intervention model. The Mean Absolute Error (MAE) of 65.41 tons indicates that the average deviation between the predicted and observed values is relatively small compared to the overall scale of monthly plastic waste generation. This result reflects a high level of agreement between the model output and the empirical data.

The Mean Absolute Percentage Error (MAPE) of 1.13% suggests that the forecasting error remains well within the range commonly classified as highly accurate in time series modeling literature. Furthermore, the Mean Percentage Error (MPE) of 0.03%, which is close to zero, implies that the model does not exhibit systematic overestimation or underestimation, indicating an unbiased forecasting structure.

The Root Mean Square Error (RMSE) of 110.09 tons, which appropriately penalizes larger deviations, remains relatively low and consistent with the MAE value. This consistency suggests that extreme prediction errors are minimal and that the model adequately captures the variability of the observed data. Overall, these accuracy measures collectively demonstrate that the model achieves an excellent statistical fit and is suitable for evaluating the impact of the implemented policy.

I. Forecasting Analysis

Figure 7 presents the 12-months forecast of monthly plastic waste volume with 95% confidence intervals. The forecast indicates that monthly plastic waste volume is expected to remain at relatively high levels throughout the forecast horizon, with clear seasonal fluctuations reflecting the strong annual periodicity.

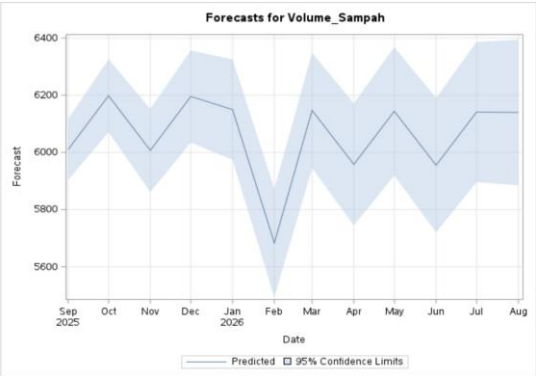

Figure 7. 12-Month Forecast of Plastic Waste Volume with 95% Confidence Intervals

J. Discussion

The counterintuitive finding that the policy intervention was associated with an increase rather than a decrease in plastic waste volume warrants careful consideration. Several explanatory mechanisms may account for this observation. First, the policy may have enhanced waste collection efficiency, thereby capturing a larger proportion of generated waste that was previously unreported or disposed through informal channels. Second, population growth and economic development during the study period may have generated upward pressure on waste volumes that exceeded any mitigating effects of the policy. Third, the policy may have been primarily focused on waste management infrastructure rather than waste reduction, potentially leading to improved documentation without corresponding reductions in waste generation [7], [8].

V. CONCLUSION

This study evaluates the effectiveness of Palu Mayor Regulation No. 40 of 2021 in reducing plastic waste generation using a Seasonal ARIMA intervention analysis. The empirical results indicate that the ARIMA(0,1,1)(1,1,1)$_{12}$ model with intervention component is the most appropriate specification (AIC = 1,321.99, MAPE = 1.13%). The intervention analysis identified a statistically significant structural change following policy implementation in March 2022, with an intervention effect of +170.40 tons per month ($p = 0.0253$), representing a 2.91% increase relative to the pre-intervention baseline.

Although the policy intervention is statistically detectable, this study cannot conclude that the regulation achieved its intended goal of reducing plastic waste during the observed period. The findings underscore the need for stronger enforcement mechanisms, complementary policy instruments, and longer-term evaluation. Future research should extend the observation period, incorporate control series for comparative analysis, and explore additional covariates to disentangle policy effects from broader socio-economic influences.

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